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Finding the MLE Transcript

Hello everyone, welcome to lesson 1 video 3. In this video, we will learn how to find the maximum likelihood estimator or MLE in practise. This is the simplest example that we can think of. Let's say you have tossed the coin 8 number of times and then you have got 8 heads, 2 tails.

Now the likelihood function in that case can be mapped to p to the power 8, 1-p, to the power 2. This is coming from binomial distribution. If you remember, you can toss a coin 10 number of times and the probability of getting a head, you can actually get the probability of getting 8 heads and 2 tails and you can map it like this. Once you have written down your likelihood function like this, now see that you have p to the power 8, 1-p square which might be inconvenient to actually maximise Lp directly.

So what we do, we take a log likelihood. So now once we have taken the log likelihood, it becomes a much more convenient function to maximise or minimise. So after this, we take a derivative of it and we find the value of p. Once we solve the value of p, we find that the value of p comes out to be 0.8. Now we can take double derivative of test and then we can find out other things as well but majorly p is equal to 0.8 if you think of was the intuition that we already had.

So if the probability of getting head is 80% or 0.8, obviously you are more inclined towards getting 8 heads in 10 trials or 800 heads in 1000 trials. That's what it means. I hope this is clear to everyone and you have got the intuition of it.

In summary, finding maximum likelihood estimator involves maximising the log likelihood function either analytically or numerically. This gives us parameter estimates that best explains the observed data. In the next video, we will learn about properties of MLEs.

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